

Data Analysis Plan and Power Justification: Aims 1 and 2

Introduction

This report provides the analysis plan and sample size justification for Aims 1 and 2 of a longitudinal study of innate immune mechanisms of cognitive decline in amnesic mild cognitive impairment (aMCI). Data are collected at baseline and one-year follow-up at the Rocky Mountain Alzheimer’s Disease Center (RMADC). Measures include peripheral plasma cytokine/chemokine panels (IL-6, TNF- α , MCP-1, Eotaxin-1), cognitive assessments, structural MRI, and Florbetapir-PET amyloid imaging (global SUVR). Six primary outcomes will be analyzed separately as one-year change scores: five memory measures and AD-signature mean cortical thickness (see Table A1 for details). The target enrollment is 137 aMCI and 55 healthy controls (HC; $N = 192$), yielding $N = 175$ after 10% attrition. A preliminary dataset ($n = 30$) was provided to inform effect size estimates.

The scientific hypotheses translate to the following statistical tests. **H1a:** A significant negative regression coefficient for baseline cytokine/chemokine level predicting one-year decline in each outcome. **H1b:** A significant negative regression coefficient for change in cytokine/chemokine level predicting the corresponding decline. **H2:** A significant cytokine $\times$ amyloid SUVR interaction coefficient, indicating steeper decline when both elevated inflammation and significant amyloid burden are present.

Methods

Data Handling. The analytic dataset will be constructed by merging blood biomarker, cognitive, neuroimaging, and demographic data from the RMADC relational database, restricted to participants who complete both baseline and follow-up visits. Distributional properties of cytokine/chemokine variables will be examined via Q-Q plots and Shapiro-Wilk tests; log-transformation will be applied only if warranted (preliminary tests suggest normality; all $p > 0.09$). Values below the detection limit will be substituted at half that

limit; outliers exceeding 4 SD will be flagged. Primary analyses will use available cases under a missing-at-random assumption; multiple imputation will serve as a sensitivity analysis.

Aim 1. Multivariable linear regression will model each one-year change score (follow-up minus baseline) as the outcome, with the baseline outcome value as a covariate to adjust for regression to the mean. For H1a, the predictor is baseline cytokine level; for H1b, the cytokine change score. Separate models will be fit for each of four markers across all six outcomes individually (see Table A1). Markers will not serve as simultaneous predictors due to extreme inter-marker correlation (preliminary IL-6/MCP-1: $r = 0.93$). Hypotheses are tested via two-sided t -tests on the cytokine regression coefficient. All models adjust for age, sex, APOE 4, BMI, hypercholesterolemia, NSAID use, immune-related conditions, and diagnostic group (aMCI vs. HC). Both groups are combined to maximize cytokine and outcome range; effect modification by diagnostic group and APOE 4 will be examined secondarily. Collinearity will be assessed via VIF (>5 triggers investigation) and the change-in-estimate criterion ($>10\%$ change warrants retention). Total cortical thickness will be included in the cortical thickness model to isolate AD-signature effects.

Aim 2. Aim 2 extends Aim 1 by adding baseline Florbetapir-PET amyloid deposition (global SUVR, the sole amyloid measure) and its product interaction with each cytokine. The primary test of H2 is the interaction coefficient, evaluated via a t -test (equivalently, an F -test with 1 numerator df for the R^2 increment). Both main effects and Aim 1 covariates are included. Secondarily, amyloid will be dichotomized ($\text{SUVR} \geq 1.10$); significant interactions will be probed via simple slopes at ± 1 SD of the cytokine, yielding a 2×2 pattern with the steepest decline predicted in the amyloid-positive/high-inflammation cell. Sensitivity analyses will stratify by APOE 4.

Multiple Comparisons. Within each of the six outcomes, FDR (Benjamini-Hochberg, $q < 0.05$) will be applied across the four markers. FDR is preferred over Bonferroni because the markers are substantially correlated (preliminary r up to 0.93), reducing the effective

number of independent tests. Bonferroni ($\alpha/4 = 0.0125$) will be a sensitivity analysis. The six outcomes will not be corrected across one another; results will be summarized by examining concordance of findings across related outcomes and markers, with consistent patterns across correlated measures interpreted as strengthening overall evidence.

Power Analysis and Sample Size Justification

We estimated power for $N = 175$ (125 aMCI + 50 HC, after 10% attrition from 192 enrolled). All tests were two-sided; power is presented at $\alpha = 0.05$ and $\alpha = 0.0125$ (Bonferroni $\alpha/4$), with the effective FDR threshold falling between these values.

Aim 1. The effect size is the Pearson correlation (r) between each marker and each outcome change score — a standardized measure of their association, estimated directly from the preliminary dataset ($n = 30$; observed $|r| = 0.26$ to 0.69). Power was estimated using `pwr.r.test()` (R `pwr` package), approximating the adjusted regression as a bivariate correlation — conservative since covariate adjustment generally increases precision. The minimum detectable $|r|$ at 80% power was 0.21 ($\alpha = 0.05$) and 0.249 ($\alpha = 0.0125$). The weakest preliminary signal (IL-6/memory, $|r| = 0.26$) achieved 94% and 84% power; all stronger associations exceeded 96% (Table 1, Figure 1A).

Aim 2. The effect size is Cohen’s $f^2 = R^2_{\text{change}} / (1 - R^2_{\text{full}})$, where R^2_{change} is the variance explained by the interaction beyond main effects and covariates. Because the preliminary data do not include amyloid SUVR, no observed interaction is available; we present the minimum detectable f^2 and power across a range of plausible values. Power was estimated using `pwr.f2.test()` (R `pwr` package; 12 predictors, numerator $df = 1$, denominator $df = 162$). The minimum detectable f^2 at 80% power was 0.048 ($\alpha = 0.05$) and 0.069 ($\alpha = 0.0125$), both small-to-medium by Cohen’s benchmarks (small = 0.02, medium = 0.15). At $f^2 = 0.05$, power was 82% ($\alpha = 0.05$); at $f^2 = 0.08$, power reached 87% at $\alpha = 0.0125$ (Table 2, Figure 1B). Assuming $R^2_{\text{full}} \approx 0.30$, the detectable interaction corresponds to $\Delta R^2 \approx 3.4\%$. The proposed N is justified by RMADC enrollment feasibility.

Tables and Figures

Table 1: Aim 1 Power: Bivariate Correlation Test ($N = 175$, two-sided). Computed using `pwr.r.test()`, R `pwr` package. Preliminary $|r|$ ranged 0.26 to 0.69.

$ r $	Power ($\alpha = 0.05$)	Power ($\alpha = 0.0125$)
0.10	0.261	0.119
0.15	0.511	0.304
0.20	0.760	0.565
0.25	0.919	0.804
0.30	0.983	0.942
0.35	0.998	0.989
0.40	1.000	0.999
0.45	1.000	1.000
0.50	1.000	1.000

Table 2: Aim 2 Power: Interaction F-test ($N = 175$, $u = 1$, $v = 162$). Computed using `pwr.f2.test()`, R `pwr` package. dR^2 approximated assuming $R^2_{\text{full}} = 0.30$. Cohen benchmarks: small = 0.02, medium = 0.15.

f^2	dR^2	Power ($\alpha = 0.05$)	Power ($\alpha = 0.0125$)
0.02	0.014	0.437	0.241
0.03	0.021	0.597	0.382
0.04	0.028	0.721	0.516

f2	dR2	Power ($\alpha = 0.05$)	Power ($\alpha = 0.0125$)
0.05	0.035	0.812	0.632
0.06	0.042	0.877	0.729
0.08	0.056	0.950	0.862
0.10	0.070	0.981	0.935
0.15	0.105	0.999	0.992

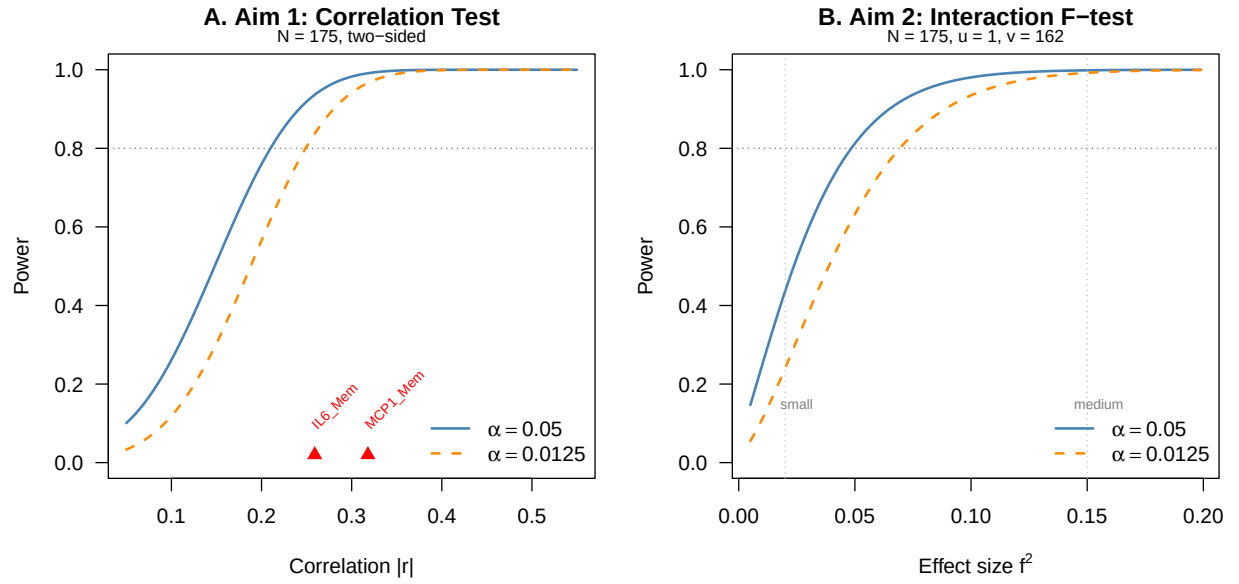


Figure 1: Power curves for Aim 1 (bivariate correlation) and Aim 2 (interaction F-test) at $N = 175$. Solid blue: $\alpha = 0.05$; dashed orange: $\alpha = 0.0125$ (Bonferroni $\alpha/4$). Red triangles in Panel A mark preliminary data correlations.

Appendix

Table A1. Variable Summary by Role

Role	Variable	Type	Measurement / Coding
Outcome 1	Change in CVLT delayed recall	Continuous	20-min delay, total correct; follow-up - baseline
Outcome 2	Change in CVLT recognition	Continuous	Recognition d-prime; follow-up - baseline
Outcome 3	Change in Benson Figure recall	Continuous	15-min delay, total correct; follow-up - baseline
Outcome 4	Change in MST pattern separation	Continuous	Proportion correct; follow-up - baseline
Outcome 5	Change in Story Recall	Continuous	1-week delay, total correct; follow-up - baseline
Outcome 6	Change in AD-signature cortical thickness	Continuous	FreeSurfer 5.1 AD-signature ROI; follow-up - baseline
Primary Predictor (Aim 1)	Plasma cy- tokines/chemokines (IL-6, TNF-a, MCP-1, Eotaxin-1)	Continuous	pg/mL; transform if warranted; H1a: baseline; H1b: change
Primary Predictor (Aim 2)	Amyloid SUVR x cytokine interaction	Continuous; Binary	Primary: SUVR x cytokine product; Secondary: amyloid-pos/neg (SUVR $\geq$ 1.10) x cytokine

Role	Variable	Type	Measurement / Coding
Covariates (All Models)	Age, sex, APOE e4, BMI, hyperc- holesterolemia, NSAID use, immune conditions, diagnostic group, baseline outcome	Continuous; Binary	Pre-specified; baseline outcome adjusts for regression to the mean

Reproducible Research

Data location: `Project_2/DataRaw/PrelimData.csv`

Code location: `Project_2/Code/Project2_PowerAnalysis.R` (standalone power analysis); `Project_2/Code/Project2_Report.Rmd` (this report)

Results location: `Project_2/Results/` (power tables as CSV; power curve figure as PDF)

GitHub repository: <https://github.com/Charl3456/BIOS6624.git>

Software: R version 4.4.1; `pwr` package (functions: `pwr.r.test()`, `pwr.f2.test()`)